

Prakhar Shah

<https://github.com/prs-016>

EDUCATION

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- **University of California San Diego** San Diego, CA
*Double Major in Computer Science and Chemical Engineering; **Provost Honors**; GPA: 3.93 Sep. 2024 – Sep. 2028*
 - **Relevant Coursework:** Advanced Data Structures, ML: Learning Algorithms, Web Mining and Recommender Systems, Discrete Math & Graph Theory, Computer Organization, Systems Programming, Statistical NLP
 - **University of Bath** Bath, England
Summer School Program for Robotics and Computer Science June 2023 – July 2023
 - **Mallya Aditi International School** Bangalore, India
High School; GPA: 4.3 June 2020 – June 2024
 - **Relevant Activities:** President:Computer Science Club, Vice-President:AI Club, Organized MathTechFest

EXPERIENCE

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- **Finiteloop** Bangalore, India
Software Engineer Intern July 2025 – Sep. 2025
 - **To-do List App:** Built and deployed a task management web app with Node.js backend and PostgreSQL database, incorporating OAuth authentication.
 - **IDS Business Next Solutions** Bangalore, India
Software Engineer Intern Nov. 2023
 - **Hotel PMS System:** Performed system-level analysis of hotel management applications to understand backend data flow and reporting modules.
 - **The Catalyst, CSforeach** La Jolla, California
Webmaster Sep. 2025 – Present
 - Helped maintain and develop the official website for *The Catalyst* and *CS foreach* organizations at UCSD by implementing responsive front-end features and improving accessibility and performance.

PROJECTS

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- **Eclipse Analytics — Project Manager**
 - Led **two concurrent sports analytics teams**, managing mentorship, sprint planning, role allocation, conflict resolution, and end-to-end delivery across multiple data-driven projects.
 - Built an ML-based optimization framework for **MLB ABS challenges** using **Statcast pitch-level data via pybaseball**, engineering spatial and contextual features (strike-zone miss encoding, pitch type, count, outs) and training an **XGBoost classifier** to predict incorrect calls and **maximize expected win probability** through challenge EV analysis.
 - Directed a parallel **NBA All-Star selection prediction project**, coordinating data preprocessing, feature engineering, and model evaluation using **historical league and player performance data**
 - **Facial Emotion Recognition System (FER2013 + YOLO + CNN + EfficientNet)**
 - Developed an end-to-end **Facial Emotion Recognition (FER)** system using the **FER2013 dataset** with a custom **CNN** and **EfficientNetB3**.
 - Implemented data preprocessing, augmentation, and fine-tuning to improve robustness.
 - Integrated **YOLOv8 face detection** and used **Grad-CAM** for explainability.
 - **FitPredict — Sentiment-Aware Recommender System**
 - Built a personalized rating predictor using **~192K real-world clothing reviews**(Rent the Runway dataset).
 - Evaluated baseline recommenders, **regularized Ridge Regression**, **Gradient Boosting**, and **FastFM** models.
 - Applied **NLP-based sentiment modeling (TF-IDF, Bag-of-Words)** to achieve a **24% RMSE reduction**.
 - Engineered hybrid features combining textual sentiment, user body measurements, item metadata, fit feedback, and temporal signals, achieving a model performance of **RMSE = 0.5436** and **MAE = 0.4124** on held-out test data.
 - **Custom Memory Allocator and UNIX Shell(C)**
 - Implemented a custom **malloc/free allocator** with **best-fit policy**, **block splitting**, **bidirectional coalescing**, and **16-byte alignment** using pointer arithmetic and bitwise metadata and a simulated **heap** manager.
 - Built a **UNIX-like (pish)shell** supporting interactive and batch execution, built-in commands, persistent history, and process management via **fork**, **execvp**, and **wait** and robust error handling using **perror**.

PROGRAMMING SKILLS

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- **Languages:** Python, C++, C, Assembly, SQL, Java, JavaScript, HTML, CSS, Rust, Go
 - **Technologies:** React, Node.js, Express, PostgreSQL, MySQL, Redis, Git, Docker, Keras, Pandas, NumPy, Matplotlib, Scikit-Learn, TensorFlow, Seaborn